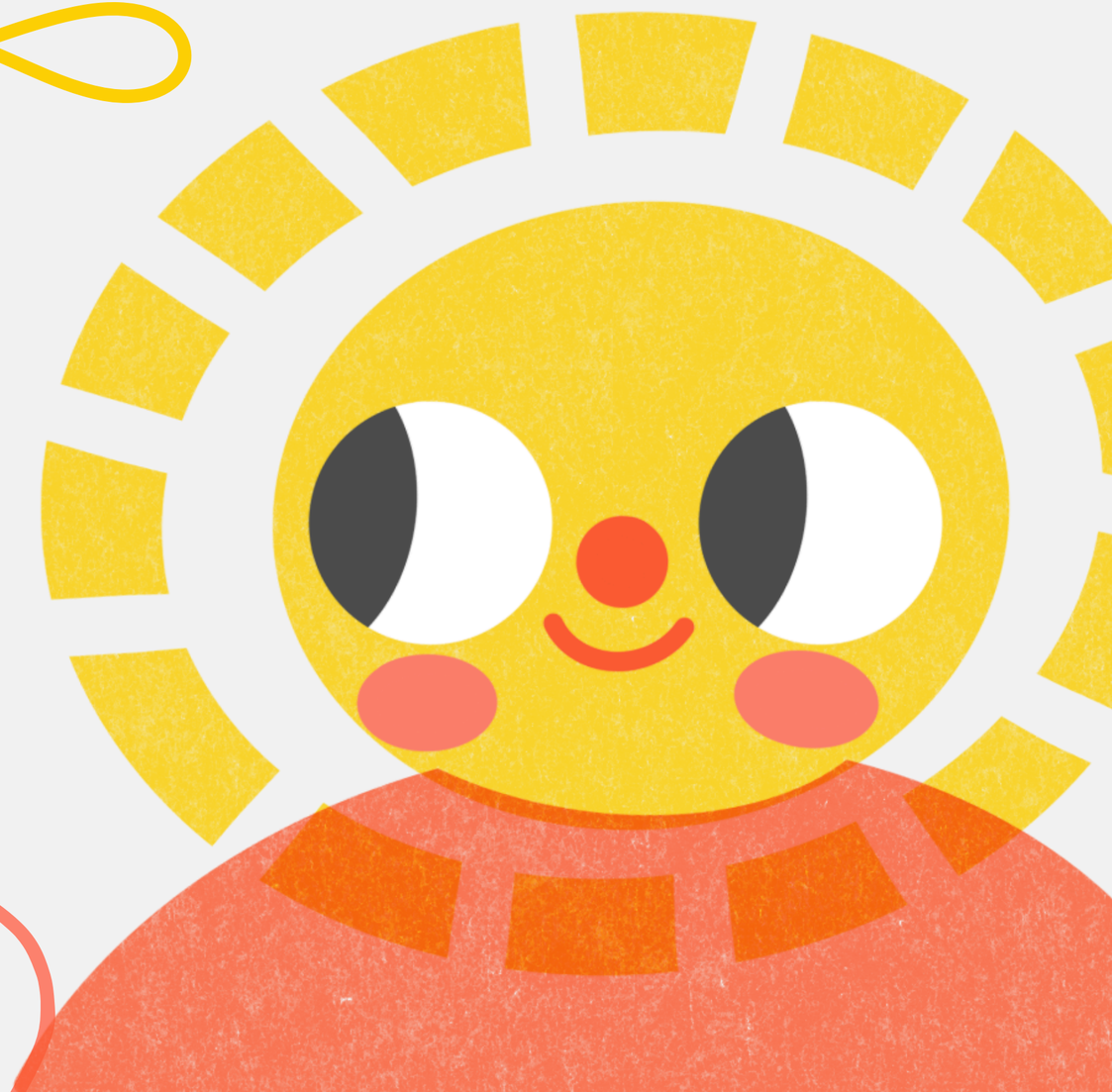
RECOGNIZE HUMAN FACIAL EXPRESSIONS

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INTRODUCTION

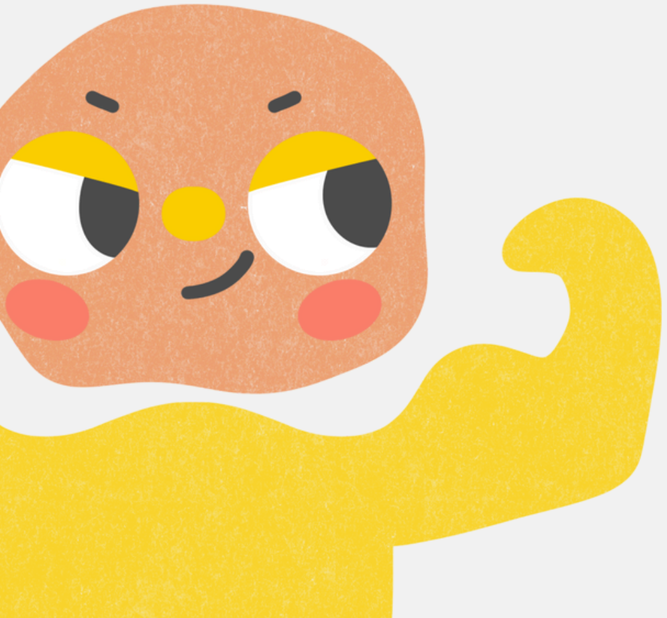
Facial expression recognition is the task of automatically identifying human emotions such as happiness, sadness, anger, and surprise from facial images. It plays an important role in human–computer interaction, healthcare, education, and security. With the power of deep learning, modern models can classify emotions with high accuracy.

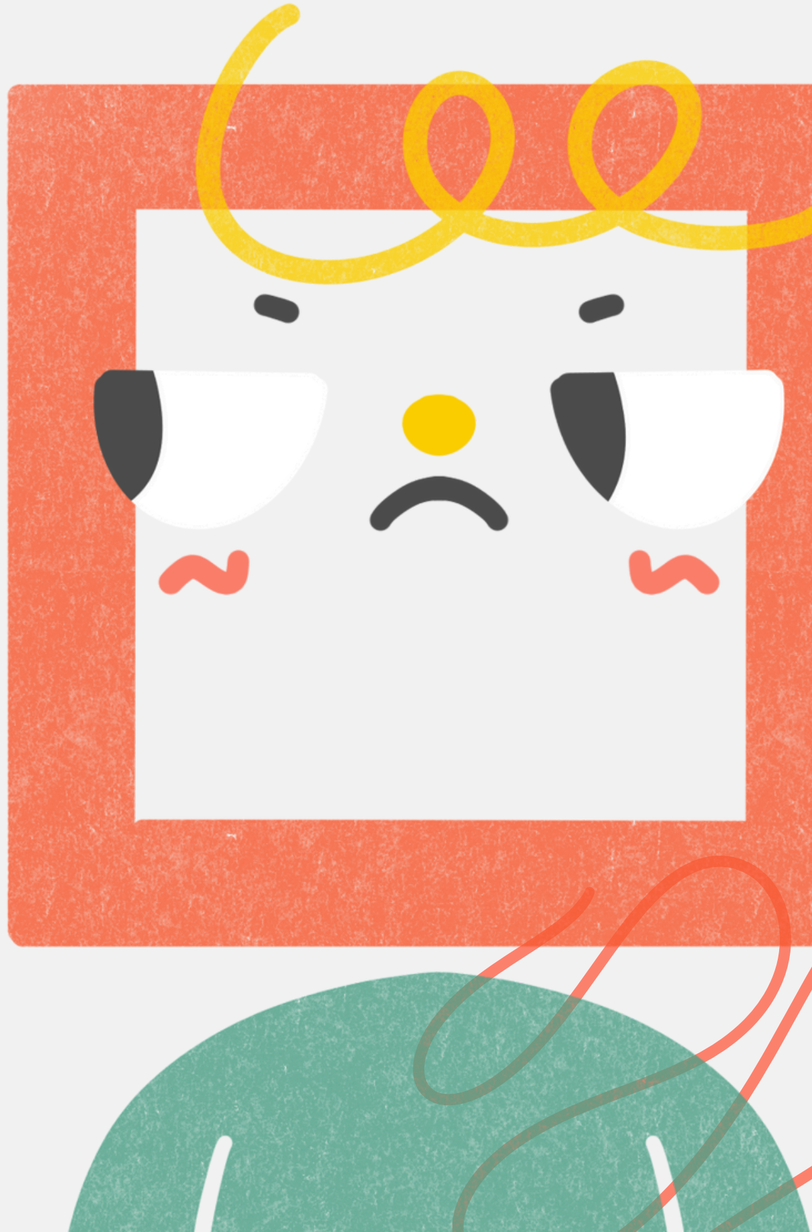


HOW CAN WE USE

Facial expression recognition can be used in many fields such as:

1. Healthcare to support patients
2. Education to measure students' engagement
3. Security for monitoring behaviors
4. Marketing or entertainment to personalize user experiences





DATASET OVERVIEW

- The dataset used is the Human Facial Expression Dataset from Kaggle.
- It contains 5 classes of facial expressions: [Happy, Sad, Angry, Surprise, and Neutral].
- The dataset is divided into train and test sets, with around 1,800+ images in total.
- Images are grayscale/colored facial photos, representing different people with various expressions.
- Data preprocessing included:
  - Resizing all images to 224 × 224 pixels.
  - Normalization (rescale = 1./255).
  - Data Augmentation (rotation, zoom, shift, flip) to increase diversity.

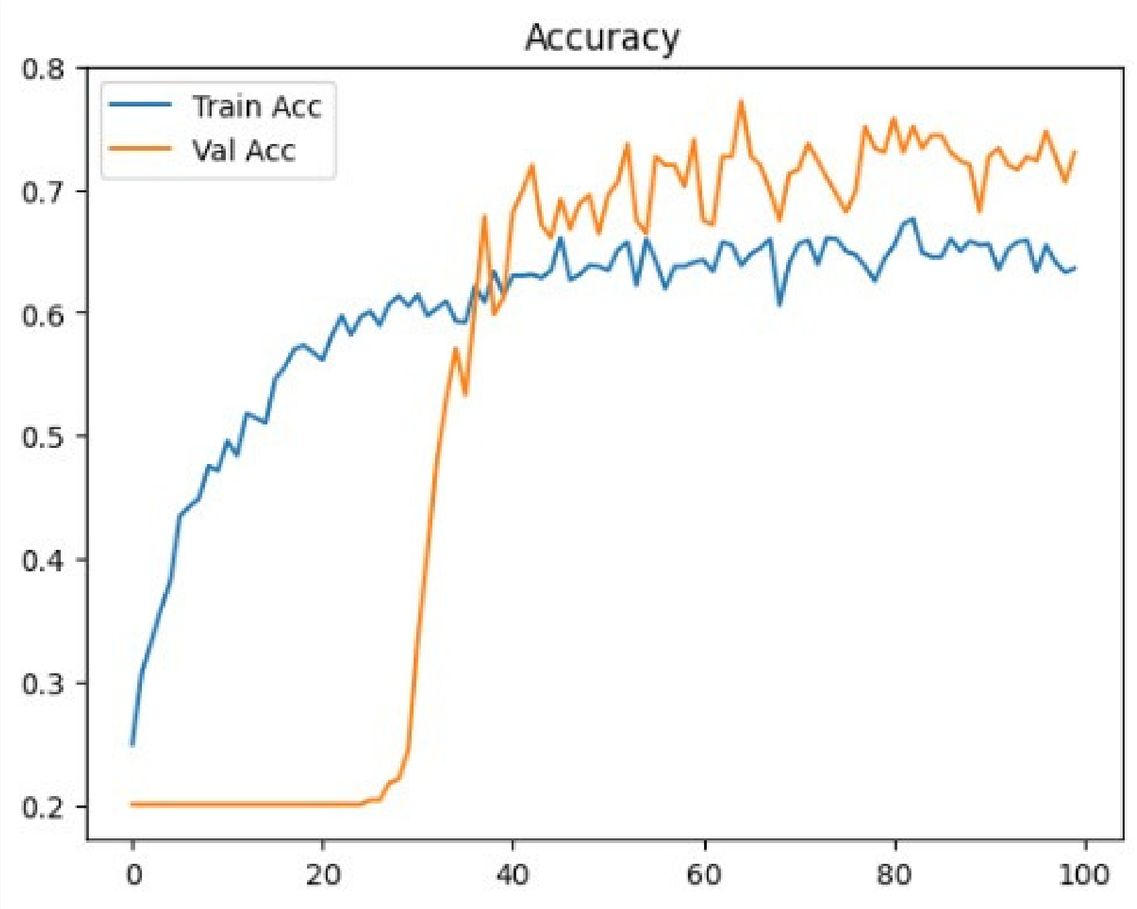


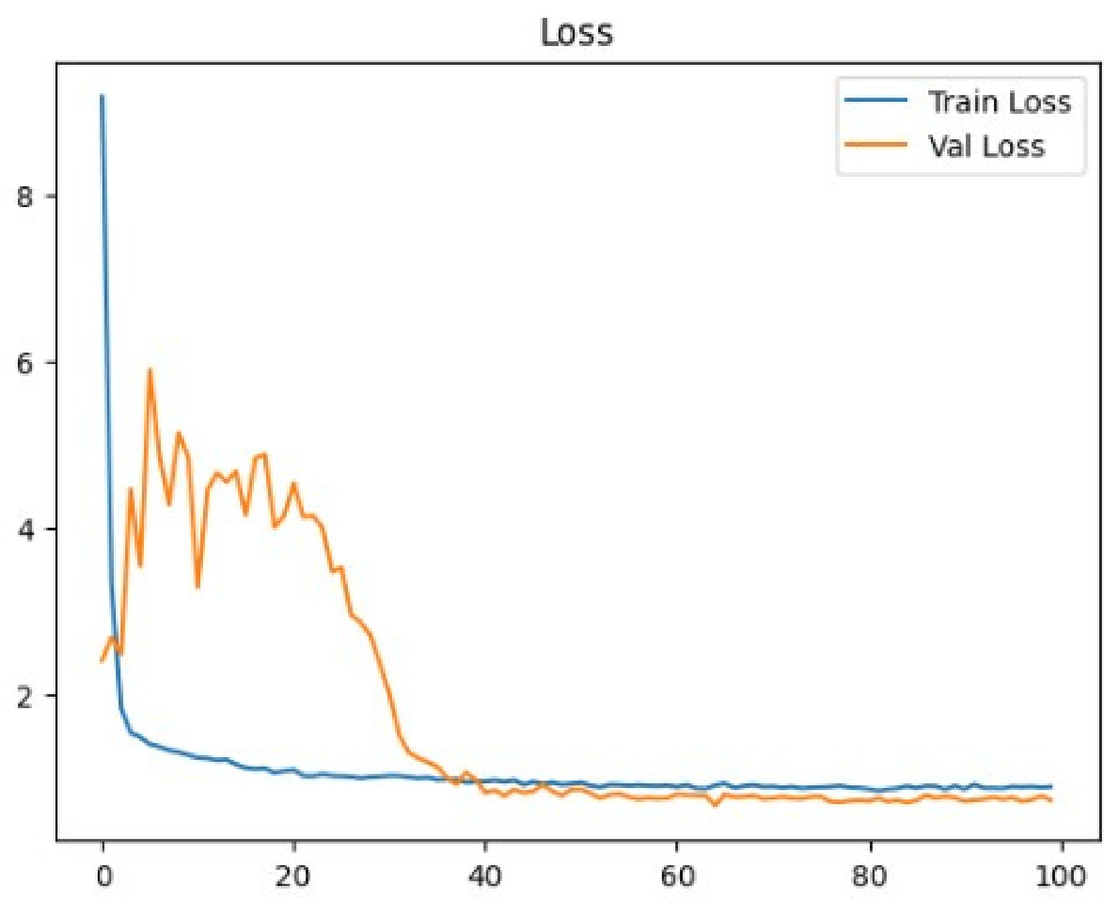
MODEL & ACCURACY OVERVIEW



- We first built a baseline CNN model from scratch, which achieved about 69% test accuracy.
- To improve performance, we applied Transfer Learning using MobileNetV2 pre-trained on ImageNet.
- With additional data augmentation and early stopping, the model reached around 80–82% accuracy on the test set.
- The results show that transfer learning significantly improves recognition of facial expressions compared to a simple CNN.

GRAPH FOR ACCURACY & LOSS



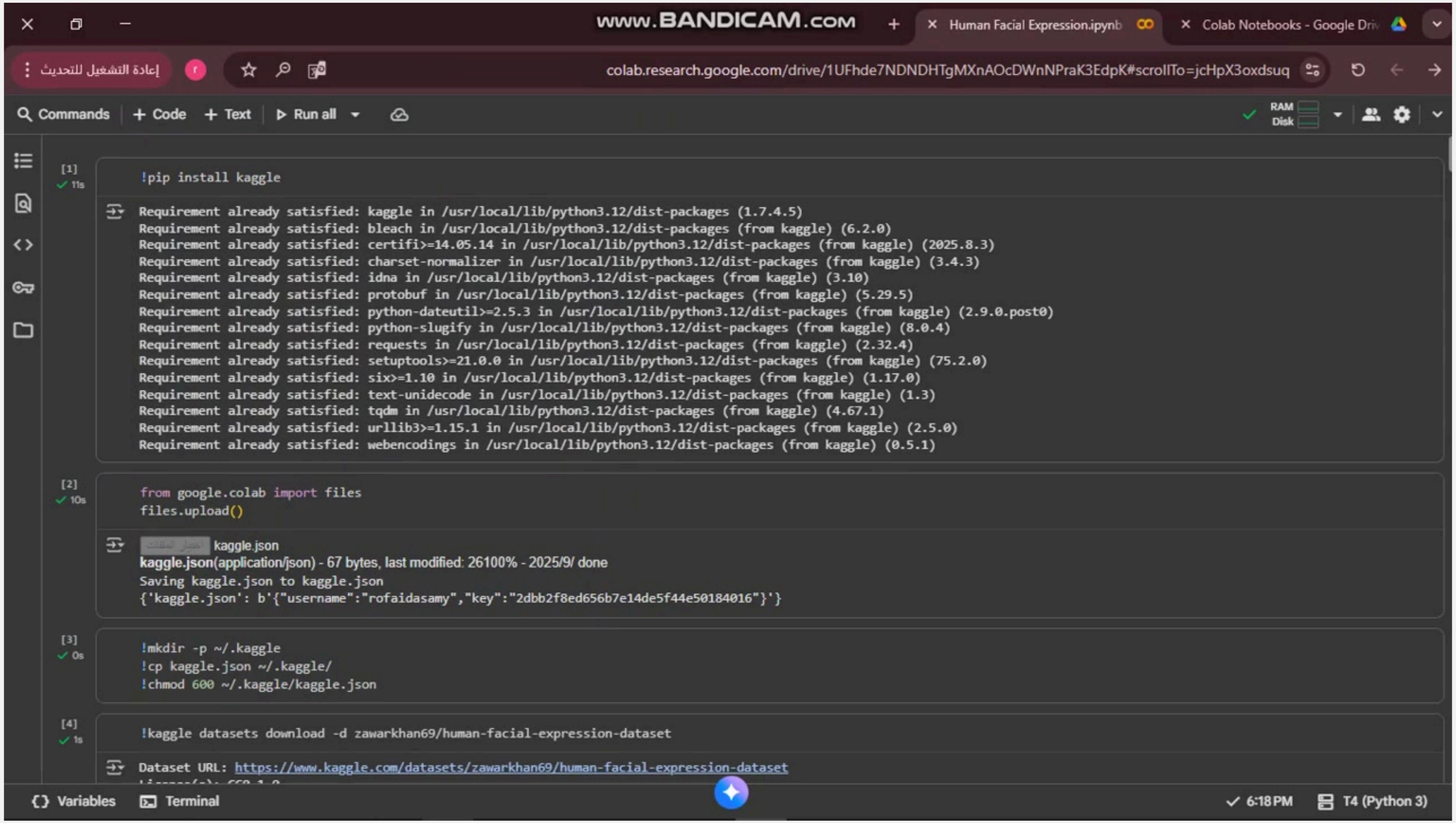


CONCLUSION

- Facial expression recognition is an important task in computer vision with many real-world applications.
- Using a simple CNN, the model reached 69% accuracy, but with transfer learning (MobileNetV2) and data augmentation, accuracy improved to 80–82%.
- This shows the effectiveness of deep learning in understanding human emotions.
- In the future, performance can be further enhanced using larger datasets, advanced models (e.g., ResNet, Vision Transformers), and real-time deployment.



VIDEO FOR THE PROJECT RUNNING



LINKS

- Dataset: https://www.kaggle.com/datasets/zawarkhan69/human-facial-expression-dataset
- Code: https://drive.google.com/file/d/1UFhde7NDNDHTgMXnAOcDWnNPraK3EdpK/view?usp=sharing
- Model: https://591d6d1410070f3c12.gradio.live


THANK YOU